

MATRIX MINDS

Data Science

SQL, Python, statistics, experimentation, and the craft of decisions.

Drafted by S. Hareedh

Founder, Matrix Minds

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Chapter 1 — What a Data Scientist Actually Does

Data Science is the practice of turning data into decisions. The job blends statistics, programming, domain knowledge, and communication. A great data scientist saves or earns the company far more than their salary by reducing uncertainty in a real decision.

The CRISP-DM cycle

Business Understanding → Data Understanding → Data Preparation → Modelling → Evaluation → Deployment. Loop forever.

Chapter 2 — SQL is Non-Negotiable

Most business data lives in relational databases. You must be fluent in SELECT, JOIN, GROUP BY, window functions (ROW_NUMBER, RANK, LAG, LEAD), CTEs, and query optimisation. Practice on LeetCode SQL and StrataScratch until it is automatic.

Window functions example

```
SELECT user_id, order_date, SUM(amount) OVER (PARTITION BY user_id ORDER BY  
order_date) AS running_total FROM orders;
```

Chapter 3 — Python for Data Work

NumPy

Vectorised arrays, broadcasting, linear algebra. The substrate everything else is built on.

Pandas

DataFrames for tabular data. Master groupby, merge, pivot_table, apply, and the difference between .loc and .iloc.

Visualisation

Matplotlib for control, Seaborn for statistical defaults, Plotly for interactivity. Every analysis ends with a chart a non-technical person can read.

Chapter 4 — Statistics You Cannot Skip

Descriptive

Mean, median, mode, variance, standard deviation, quartiles, IQR. Always look at the distribution before computing summary statistics.

Inferential

Confidence intervals, hypothesis testing, p-values (and their misuse), t-tests, chi-squared, ANOVA. A p-value is not the probability the null is true.

Bayesian thinking

Update beliefs proportionally to evidence. Useful for A/B tests with small samples and for combining priors with data.

Chapter 5 — Exploratory Data Analysis (EDA)

Before any modelling: profile every column (dtype, missing %, unique values, distribution), look for outliers, check class balance, and form hypotheses about relationships. Tools: pandas-profiling, sweetviz, or a disciplined notebook.

Common traps

Survivorship bias, Simpson's paradox, look-ahead bias in time series, and selection bias from how the data was collected.

Chapter 6 — Experimentation and A/B Testing

Randomised controlled experiments are the gold standard for causal claims. Define one primary metric, compute sample size in advance, run the experiment until that size is reached, then read the result once.

Pitfalls

Peeking (early stopping inflates false positives), novelty effects, network effects between users, and chasing statistically significant but practically tiny lifts.

Chapter 7 — Communicating Results

The output of data science is not a notebook — it is a decision. Lead with the recommendation, then the evidence, then the caveats. Use one chart per point. Quantify uncertainty.

Dashboards

Build dashboards for recurring questions (Looker, Metabase, Superset). Build bespoke analyses for one-off decisions.

Chapter 8 — Your First Data Science Portfolio

Hiring managers want evidence, not certificates. Publish 3 projects on GitHub: (1) an end-to-end analysis with a clear business question, (2) a predictive model with proper evaluation, (3) a dashboard or small data product. Write a README that a non-technical reader can understand.

Closing

Curiosity plus rigour beats credentials. — S. Hareedh, Matrix Minds

Thank you for reading.

This eBook was drafted by **S. Hareedh**, Founder of Matrix Minds. If it helped you, share it with one friend who is starting out. For feedback, write to **matrixmindsha@gmail.com**.